

Shubham Pakale

Software Developer

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PROFESSIONAL SUMMARY

Software Developer with 2+ years of experience building production AI systems and scalable backend platforms. At Simplify Healthcare, designs and ships GenAI-powered RAG pipelines, integrates LLMs into healthcare workflows, and modernizes .NET systems at scale. Independently architected an agentic DSA chatbot with intelligent query routing, a custom MCP server exposing PostgreSQL via natural language, and an async Pinecone embedding pipeline — demonstrating full ownership of AI system design from data ingestion to deployed inference. Core stack: .NET Core, Python, ReactJS, LangChain, Azure OpenAI, PostgreSQL, and Pinecone.

EDUCATION

Bachelor of Technology in Information Technology, <i>Pimpri Chinchwad College of Engineering, Pune, India</i> CGPA: 8.0/10 Relevant Courses: Operating Systems, Data Structures, Algorithms, Artificial Intelligence, Machine Learning, Networking, Databases	12/2021 – 07/2024 Pune, India
Diploma in Information Technology, <i>Puranmal Lahoti Government Polytechnic, Latur, India</i> Percentage: 95.6%	07/2018 – 07/2021 Latur, India

SKILLS

Languages C#, Python, JavaScript, SQL, C++, Java, Go, Bash	Backend Frameworks & Frontend: .NET Core, ASP.NET MVC, Entity Framework, Dapper, FastAPI, Hangfire, ReactJS, RESTful API Integration
AI / ML LangChain, Google Gemini API, Azure OpenAI, OpenAI API, RAG Pipelines, Prompt Engineering, MCP (Model Context Protocol), FastMCP	Architecture Layered Architecture, Dependency Injection, Multi-tenancy, Async Programming
Vector & Relational Databases Pinecone, FAISS, PostgreSQL, SQL Server, MySQL	DevOps & Cloud Azure, AWS, Docker, Git, CI/CD Pipelines, Ansible, Linux

PROFESSIONAL EXPERIENCE

Associate Software Developer, Simplify Healthcare - Built and maintained scalable healthcare platforms using layered architecture and dependency injection in .NET Core, reducing technical debt by 25% and improving system reliability by 30%. - LLM-Powered Healthcare Chatbot: Designed and deployed a RAG pipeline using Azure OpenAI with PostgreSQL as the retrieval backend — used RecursiveCharacterTextSplitter for chunking, cosine similarity for retrieval, and prompt templating for grounded responses; improved query relevance by 25%. - Designed and optimized SQL Server and PostgreSQL schemas and queries using Dapper and Entity Framework, reducing data access time by 35%.	10/2024 – Present Pune, India
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- Migrated legacy applications from .NET Framework to .NET Core, improving maintainability and long-term reliability.
- Developed multi-tenant application framework, cutting client onboarding time by 50%.
- Deployed Azure cloud solutions using Azure Functions and Hangfire for background task automation.

Software Engineer Intern, Zscaler Inc.

03/2024 – 09/2024

Pune, India

- Collaborated on software development, testing, and automation projects, receiving excellent performance feedback.
- Executed DevOps tasks including CI/CD pipelines, Ansible and Bash scripting, reducing server downtime by 25% and deployment time by 45%.
- Created Ansible playbooks to configure Linux nodes for Zscaler cloud integration.
- Automated testing with PyATS, developing 20+ test cases, enhancing quality assurance.

PROJECTS

Intelligent DSA Q&A System,

| Python, LangChain, RAG, Pinecone, PostgreSQL, MCP, Google Gemini API, FastMCP

- Architected a full-stack AI chatbot with a smart routing engine that directs questions to the optimal backend RAG (Pinecone vector search) for theory/concept queries, MCP+PostgreSQL for structured data queries, or a hybrid of both.
- Built an async PDF ingestion pipeline using LangChain + PyPDFLoader, chunking DSA content and embedding it via Google text-embedding-004 into Pinecone, enabling semantic similarity search across the knowledge base.
- Implemented a custom MCP (Model Context Protocol) server using FastMCP that exposes a PostgreSQL tool; integrated it as an stdio subprocess client, with Gemini dynamically generating and executing SQL from natural language queries.
- Added multi-turn conversation memory, model fallback chain across Gemini Flash/Pro variants, and auto-retry SQL validation — achieving 95% query success rate and 60% faster information retrieval.

Skin Disease Classification (BTech Final Year Project)

- Developed a system for classifying skin diseases using deep learning and image processing techniques, training 10 CNN models on the HAM10000 dataset.
- Enhanced image classification accuracy by integrating Convolutional Neural Networks (CNN) with Gray-Level Co-Occurrence Matrix (GLCM) features, achieving a 15% increase in precision and recall metrics in medical imaging applications.
- Applied Support Vector Machine (SVM) for final classification, achieving 98% accuracy and demonstrating proficiency in machine learning and image processing.

HONORS & AWARDS

Top 20 Rank & Seva Sahayog Student Scholarship

1st Place, WebXplore Hackathon - NSCC

Top 100 Achievers, UNO MINDS 3.0